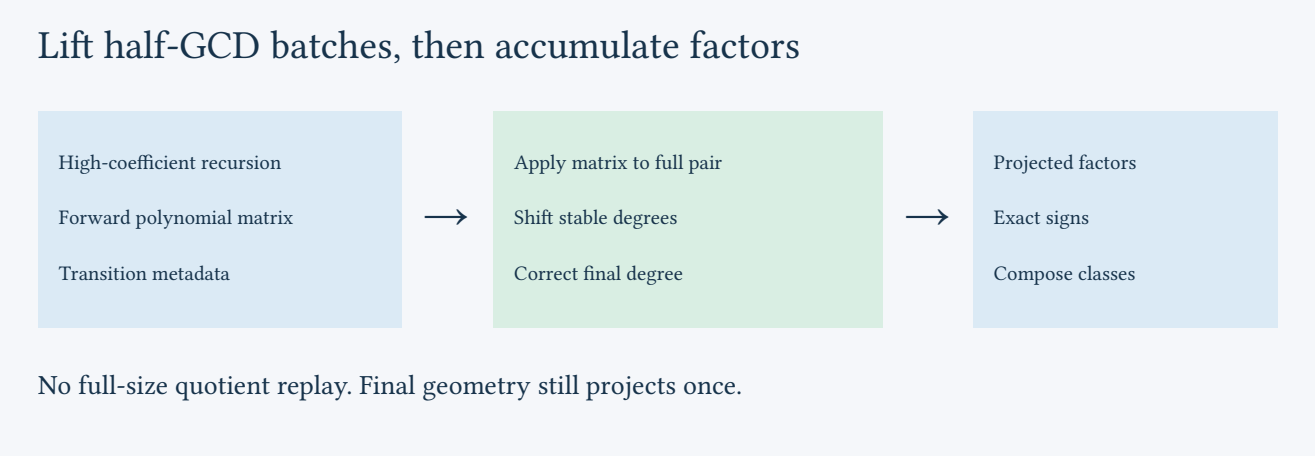
Half-GCD projected-resultant batches

Exact lifted transition factors for the prime-field source



Construction

Recursively compute a forward polynomial matrix from high coefficients, apply it to the full input pair, and compose the next batch. Store each transition's dividend/divisor/remainder degrees and leading divisor coefficient. Shift stable degrees on lifting and replace the final remainder degree with its actual lifted degree before accumulating resultant factors.

Exact checks

960 cases match a separate schoolbook Euclidean trace, including every transition, returned pair, matrix action/determinant and the projected resultant. The panel exercises 1679 recursive calls and 17 changed-final-degree lifts. Geometry exercises 174 recursive calls; all 2500 cases retain agreement and exactly one projection per valid case. Eleven selected native methods pass. Generator replay is exact.

Cost boundary

The implementation applies polynomial matrices rather than replaying full-size divisions. Matrix products, allocations and division work remain. The degree-eight cutoff is not tuned; no elapsed-time advantage has been measured against either earlier backend. Geometry point-list and decoder-construction costs remain.

Evidence and next step

Native logs, compiler warnings, binary/dependency hashes and commands are preserved. Checks are producer-authored. Next: complete polynomial-work accounting against both references, then implicit orbit products and compact nonlinear decoder construction. No cloud or recovery comparison ran.

Source: FLINT half-GCD lifting/resultant documentation, linked with the derivation in DERIVATION.md. KB unavailable. Previous evidence and original cohort unchanged.